

Chapter 9: Network Management and Security

What is Network Management?

"Network management includes the deployment, integration and coordination of the hardware, software, and human elements to monitor, test, poll, configure, analyze, evaluate, and control the network and element resources to meet the real-time, operational performance, and Quality of Service requirements at a reasonable cost."

ISO Network Management Areas

The International Standards Organization (ISO) has defined a network management model and framework referred to as FCAPS (fault, configuration, accounting, performance, security).

- **Fault Management** : monitoring and management of faults that occur in a network
- **Configuration Management** : management of network device configurations and changes
- **Accounting Management** : billing management
- **Performance Management** : monitoring and management of the efficiency or utilization of the network
- **Security Management** : controlling access to the network and network devices

SNMP (Simple Network Management Protocol)

It is an application layer protocol used to monitor network-attached devices. It is part of the TCP/IP protocol suite. It allows Administrators to manage network performance, find and solve network problems, and plan for network growth. SNMP works as the manager/agent model: Manager (the monitoring "client") and Agent (running on the equipment server). The manager and agent use a Management Information Base (MIB) and a relatively small set of commands to exchange information. MIBs are files defining the objects that can be queried, including: Object name, Object description and Data type (integer, text, list). The MIB is organized in a tree structure with individual variables, represented as leaves on the branches. A long numeric tag or object identifier (OID) is used to distinguish each variable uniquely in the MIB and in SNMP messages.

How does it work?

Basic commands

GET (manager -> agent) - Query for a value

GET-NEXT (manager -> agent) - Get next value (list of values for a table)

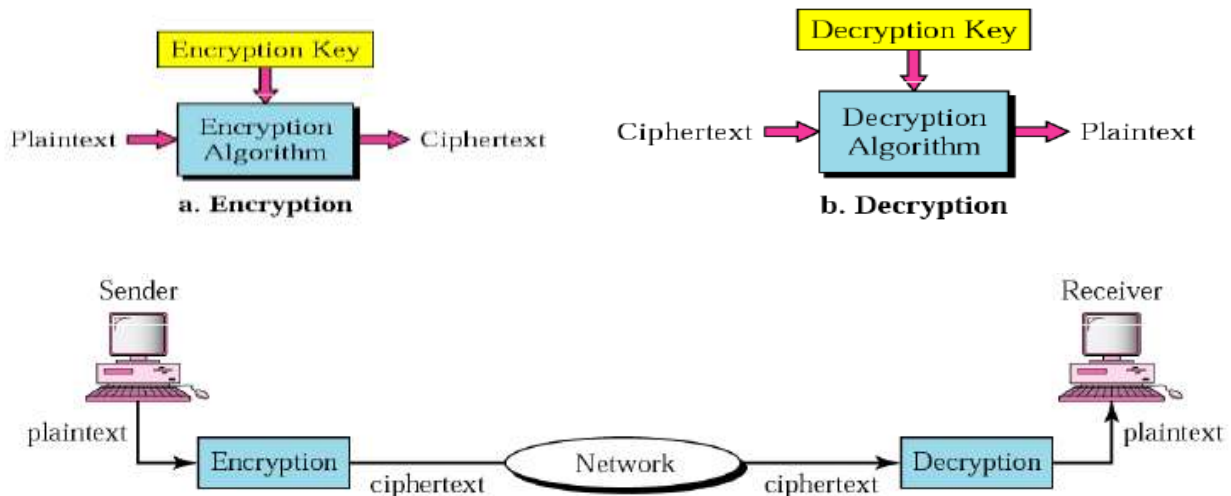
GET-RESPONSE (agent -> manager) - Response to GET/SET, or error

SET (manager -> agent) - Set a value, or perform action

TRAP (agent -> manager) - Spontaneous notification from equipment (line down, temperature above threshold, ...)

Cryptography

Cryptography in Greek means “Secret Writing”. It is science and art of transforming message to make them secure and immune to attack. Original message before transformation is Plaintext. An Encryption algorithm transforms Plaintext to Cipher-text. Decryption algorithm transforms Cipher-text to Plaintext. Cipher refers to different categories of algorithm in Cryptography.

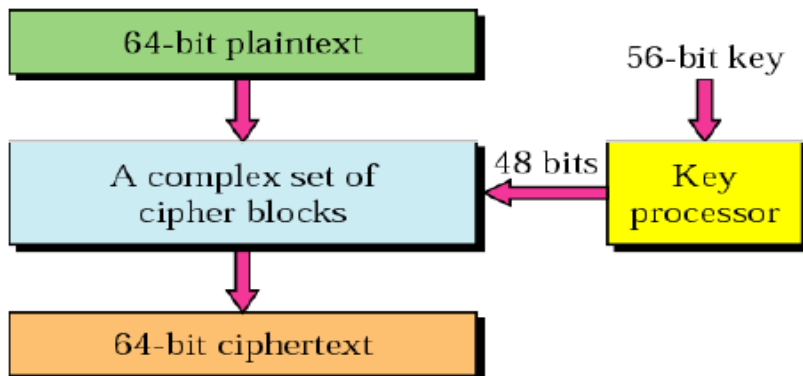


Symmetric key cryptography: sender, receiver keys *identical*

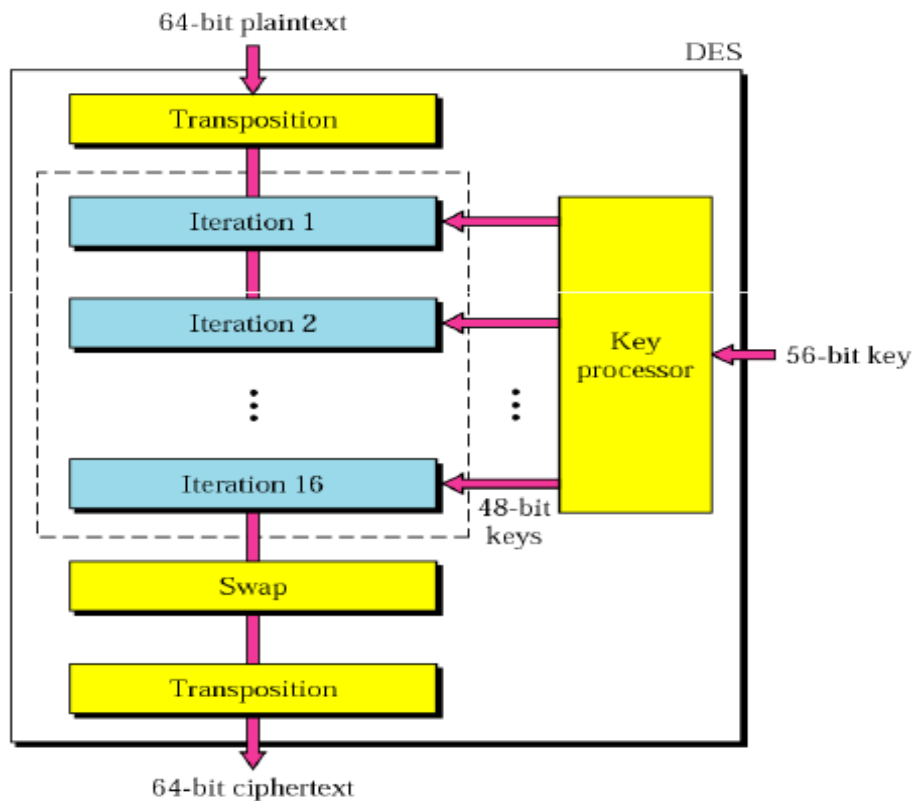
Public-key cryptography: encryption key *public*, decryption key *secret* (private)

Symmetric key cryptography: DES (Data Encryption Standard)

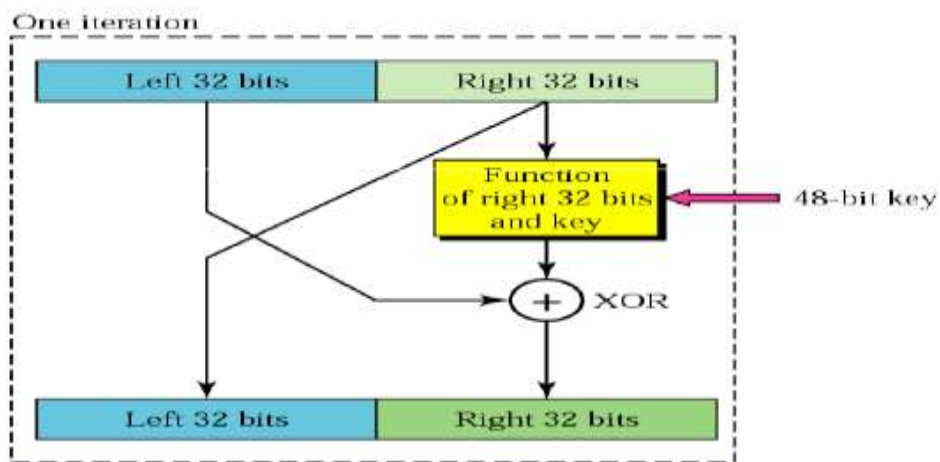
The cryptographic algorithm specified in this standard transforms a 64-bit binary value into a unique 64-bit binary value based on a 56-bit variable. There are 72,000,000,000,000,000 (72 quadrillion) or more possible encryption keys that can be used



DES General Scheme



DES iteration block



During Encryption

$$L_n = R_{n-1}$$

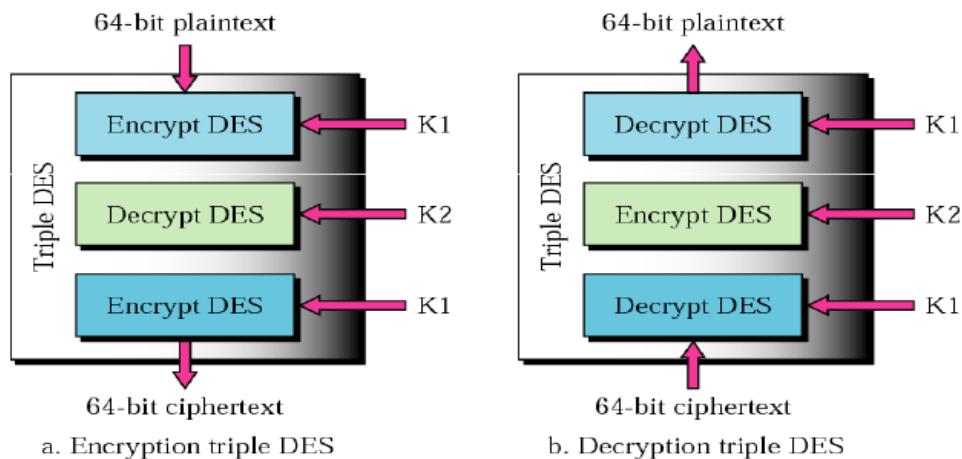
$$R_n = L_{n-1} \oplus f(R_{n-1}, K_n)$$

During Decryption

$$R_{n-1} = L_n$$

$$L_{n-1} = R_n \oplus f(L_n, K_n)$$

Triple DES



Public Key Cryptography: RSA

It was invented by Ron Rivest, Adi Shamir and Leonard Adleman . It supports Encryption and Digital Signatures. It is most widely used Public Key Algorithm. It gets its security from integer factorization problem. It is relatively easy to understand and implement. RSA is used in Security Protocols such as- IPSEC/IKE => IP Data Security, SSH => Terminal Connection Security, TLS/SSL => Transport Data Security

RSA Algorithm : Key Generation

1. Generate Two Large Prime Numbers, p and q
2. Let $n = pq$
3. Let $m = (p-1)(q-1)$
4. Choose a small number e ($1 < e < m$), co prime to m such that $GCD(e, m) = 1$.
5. Find d , such that $de \% m = 1$. Where $d = (1 + m * i)/e$
6. Publish e and n as the Public Key
7. Keep d and n as the Private Key

RSA example

Key Generation: Steps

1. Generate Two Large Prime Numbers, p and q
 $Let\ p = 7\ and\ q = 19$
2. Let $n = pq$
 $n = 7 * 19$
 $n = 133$

3. Let $m = (p-1)(q-1)$
 $m = (7-1)(19-1)$
 $m = 6 * 18$
 $m = 108$
4. Choose a small number e ($1 < e < m$), co prime to m such that $GCD(e, m) = 1$
 $e = 2 \Rightarrow GCD(e, 108) = 2$ (no)
 $e = 3 \Rightarrow GCD(e, 108) = 3$ (no)
 $e = 4 \Rightarrow GCD(e, 108) = 4$ (no)
 $e = 5 \Rightarrow GCD(e, 108) = 1$ (yes!) $\Rightarrow GCD(e, m) = 1$
5. Find d , such that $de \% m = 1$. Where $d = (1 + m * i)/e$
 [Go through Values of i until Integer Solution is Found]
 $i = 0 \Rightarrow d = 1 / 5$ (No Integer)
 $i = 1 \Rightarrow d = 109 / 5$ (No Integer)
 $i = 2 \Rightarrow d = 217 / 5$ (No Integer)
 $i = 3 \Rightarrow d = 325 / 5 = 65$ (Yes !! Satisfies the Condition)
6. $(n, e) = (133, 5)$ AND $(n, d) = (133, 65)$

RSA: Encryption, decryption

Given (n, e) and (n, d) as computed above

To encrypt bit pattern m , compute

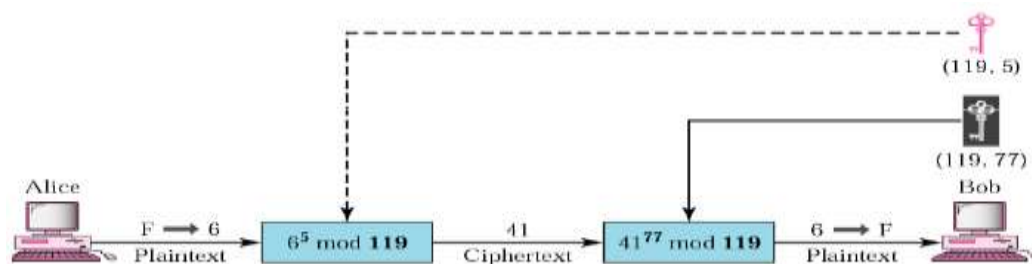
$$c = m^e \bmod n \quad (\text{i.e., remainder when } m^e \text{ is divided by } n)$$

To decrypt received bit pattern, c , compute

$$m = c^d \bmod n \quad (\text{i.e., remainder when } c^d \text{ is divided by } n)$$

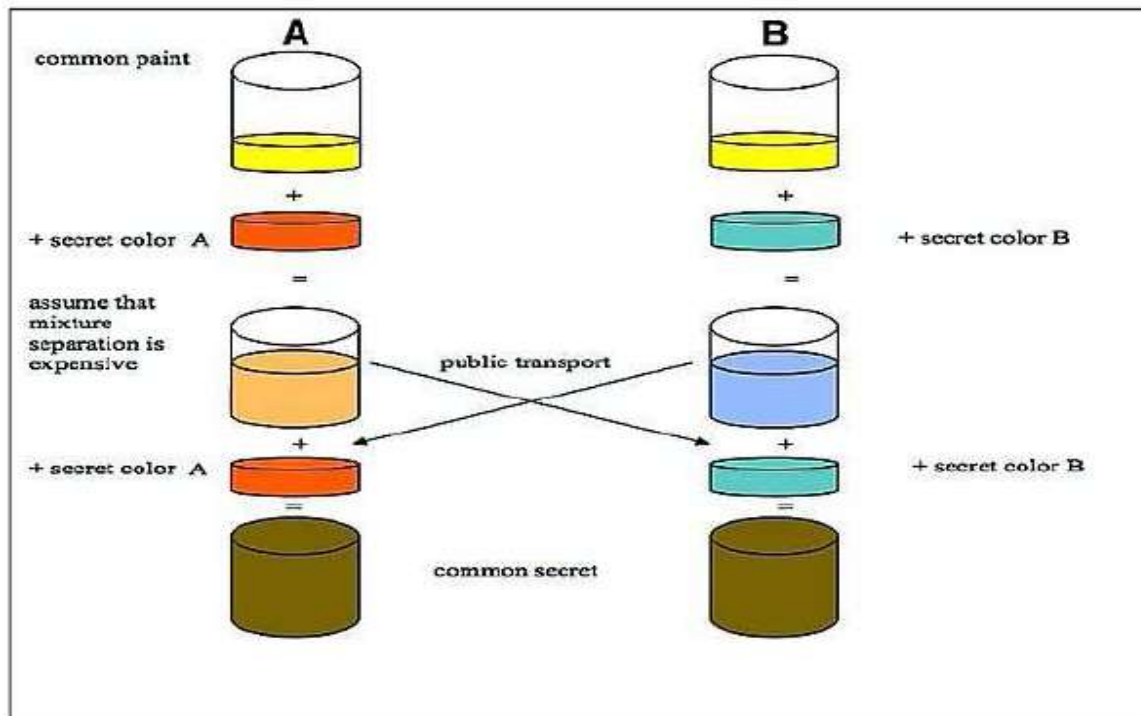
RSA : Encryption, Decryption Example

- Public Key Pair $\Rightarrow (n, e) \Rightarrow (119, 5)$
- Private Key Pair $\Rightarrow (n, d) \Rightarrow (119, 77)$



Key Exchange Protocol: Diffie Hellman Algorithm

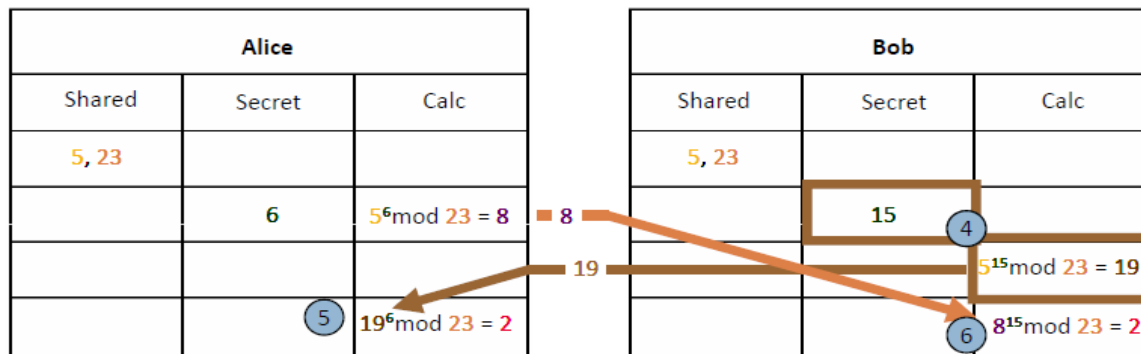
Analogy



Diffie Hellman Algorithm : Step by Step Illustration

Alice			Bob		
Shared	Secret	Calc	Shared	Secret	Calc
1 5, 23			1 5, 23		
	2 6	3 $5^6 \text{ mod } 23 = 8$			

1. Alice and Bob agree to use the same two numbers.
For example, the Base Number $g=5$ and Prime Number $P=23$
2. Alice now chooses a Secret Number $x=6$.
3. Alice performs the DH algorithm: $g^x \text{ MOD } P = (5^6 \text{ modulo } 23) = 8$ (Y)
and sends the New number 8 (Y) to Bob



4. Meanwhile Bob has also chosen a Secret Number $x=15$, performed the DH algorithm: $g^x \text{ modulo } P = (5^{15} \text{ MOD } 23) = 19$ (Y) and sent the new number 19(Y) to Alice.

5. Alice now computes $Y^x \text{ modulo } P = (19^6 \text{ MOD } 23) = 2$.

6. Bob now computes $Y^x \text{ modulo } P = (8^{15} \text{ MOD } 23) = 2$.

The Result (2) is the same for both Alice and Bob. This number can now be used as a shared secret key by the encryption algorithm.

Kerberos

In Greek Mythology, *Kerberos* (or *Cerberus*) is the Horrible three-headed guard dog of Hades.

Hades was the Ancient Greek god of the Underworld. The Underworld is a region that is thought to be under the surface of the Earth.

Kerberos is an authentication service developed as part of Project Athena. It is one of the best known and most widely implemented **Trusted Third Party** key Distribution Systems.

Designed to Provide Strong Authentication for Client/Server Applications by using Secret-Key Cryptography. It uses **Strong Cryptography** so that a Client can Prove its Identity to a Server (and vice versa) across an Insecure Network Connection. Two Versions of Kerberos are in Common Use: v4 & v5.

IPSec

IPSec is a framework of open standard for ensuring private communication over public network. It has become the most common network layer security control, typically used to create a VPN. A VPN is a virtual network built on top of existing physical networks that can provide a secure communications for data transfer. VPNs are used most often to protect communications carried over public networks such as the Internet. A VPN can provide several types of data protection: confidentiality, integrity, data origin authentication, replay protection and access control.

IPSec Components :

Two security protocols -Authentication Header (AH) and Encapsulating Security Payload (ESP). AH can provide integrity protection for packet headers and data, but it cannot encrypt them. ESP can provide encryption and integrity protection for packets, but it cannot protect the outermost IP header, as AH can. However, this protection is not needed in most cases. Accordingly, ESP is used much more frequently than AH because of its encryption capabilities.

Internet Key Exchange (IKE) protocol:

IPsec uses IKE to negotiate IPsec connection settings. It authenticates endpoints to each other. It defines the security parameters of IPsec-protected connections. It negotiates secret keys and manages, updates, and deletes IPsec-protected communication channels.

IP Payload Compression Protocol (IPComp):

Optionally, IPsec can use IPComp to compress packet payloads before encrypting them.

IPSec Mode

Transport mode

IPSec header is inserted just after the IP header. The Protocol field in the IP header is changed to indicate that an IPSec header follows the normal IP header (before the TCP header). The IPSec header contains security information

Tunnel mode

AH or ESP fields are added to the IP packets, the entire packet. Tunnel mode is used when both ends or one end uses a firewall or router that implements IPsec. The unprotected packets can pass through external networks. Tunnel mode is also useful when the tunnel ends at a location other than the final destination. The disadvantage of tunnel mode is that it adds an extra IP header, thus increasing packet size substantially. In contrast, transport mode does not affect packet size.

Benefits of IPSec

- It provides strong security
- IPSec in a firewall is resistant to bypass if all traffic from outside sees IP
- IPSec is below transport layer and is transparent to applications
- IPSec can be transparent to end users
- IPSec can provide security to individual users if needed

Applications of IPSec

- Secure branch office connectivity over the internet
- Secure remote access over the internet
- Establishing extranet and intranet connectivity with partners

VPN (Virtual Private Network)

A VPN is a virtual network built on top of existing physical networks. It provides a secure communications mechanism for data and IP information transmitted between networks. VPN can be used over existing networks, such as the Internet. Thus, it can facilitate the secure transfer of sensitive data across public networks. This is often less expensive than alternatives such as dedicated private telecommunications lines between organizations or branch offices. Firewalls, VPNs, and IPsec with ESP in tunnel mode are a natural combination and widely used in practice. VPNs can use both symmetric and asymmetric forms of cryptography.

Three primary models for VPN architectures :

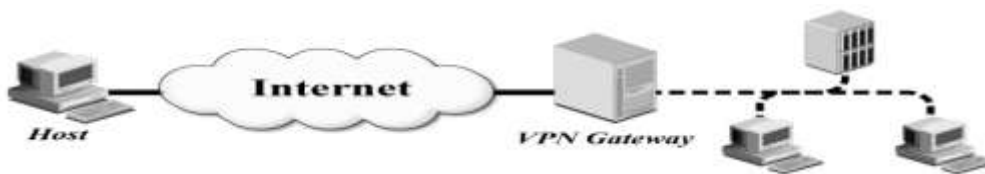
Gateway-to-gateway:

- Protects communications between two specific networks
- Eg., an organization's main office network and a branch office network, or two business partners networks.



Host-to-gateway:

- Protects communications between one or more individual hosts and a specific network belonging to an organization.
- Eg., traveling employees to gain access to internal organizational services, such as the organization's e-mail and Web servers.



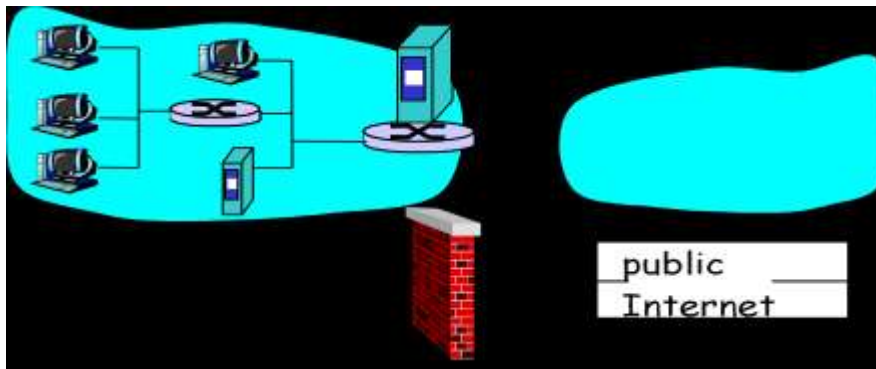
Host-to-Host :

- Protects communication between two specific computers.
- Eg., small number of users need to use or administer a remote system



Firewall

It Isolates organization's internal network from larger Internet, allowing some packets to pass, blocking others. It acts as a security gateway between two networks. A Firewall defines a single choke point that keeps unauthorized users out of the protected network, prohibits potentially vulnerable services from entering or leaving the network, and provides protection from various kinds of IP spoofing and routing attacks. The use of single choke point simplifies security management because security capabilities are consolidate on a single system. Windows Firewall helps protecting your computer by preventing unauthorized users from gaining access to your computer through a network or internet.



Firewalls: Why

- It prevents denial of service attacks: SYN flooding: attacker establishes many bogus TCP connections, no resources left for “real” connections.
- It prevents illegal modification/access of internal data. e.g., attacker replaces CIA's homepage with something else
- It allow only authorized access to inside network (set of authenticated users/hosts)
- It preserves customer and partner confidence: prevent viruses and worms on your network and prevent malicious attackers from getting into your network

Firewall : Types of Firewall

1. Packet Filtering Router

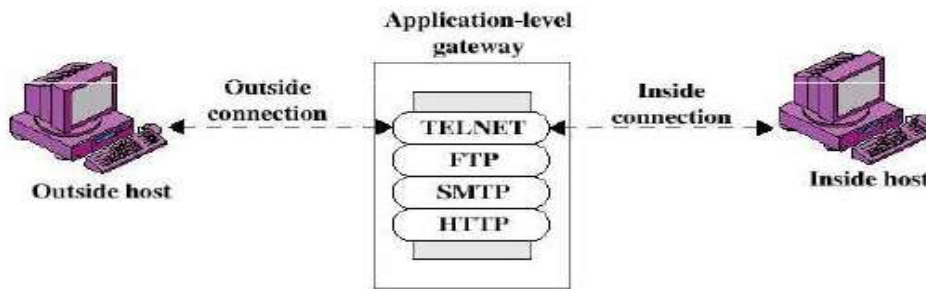
It applies a set of rules to each incoming IP Packet. The router is configured to filter packets going in both directions. Router filters packet-by-packet, decision to forward/drop packet based on: source IP address, destination IP address, TCP/UDP source and destination port numbers, ICMP message type, TCP SYN and ACK bits

Example : block incoming and outgoing datagram with IP protocol field = 17 and with either source or dest port = 23.

All incoming and outgoing UDP flows and telnet connections are blocked.

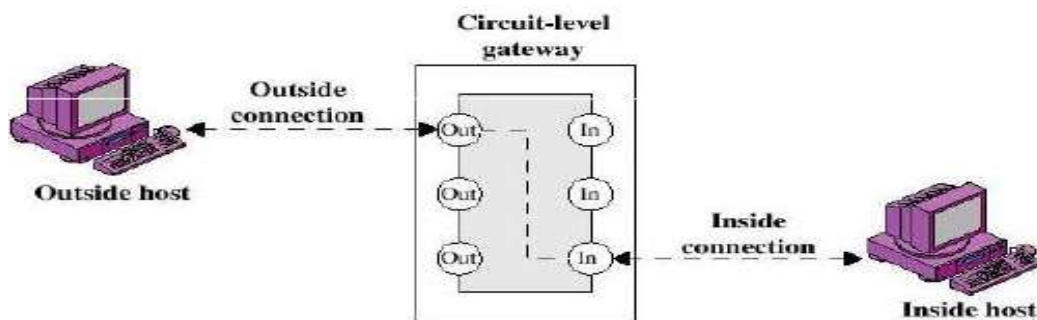
2. Application Level Gateway

They are called Proxy Servers and acts as a relay of application level traffic. Example: allow selected internal users to telnet outside. Require all telnet users to telnet through gateway. For authorized users, gateway sets up telnet connection to destination host. Gateway relays data between 2 connections. Router filter blocks all telnet connections not originating from gateway.



3. Circuit Level Gateway

It does not permit an end to end TCP Connection directly. The gateway setups two TCP Connections (IN and OUT). It monitor TCP handshaking between packets to determine whether a requested session is legitimate. Once two connections are established => Gateway Relays



Limitations of firewalls and gateways

- IP spoofing: router can't know if data "really" comes from claimed source
- The firewall cannot protect against attacks that bypass the firewall.
 - Internal systems may have dial-out capability to connect to an ISP.
- The firewall does not protect against internal threats, such as a dishonest employee or an employee who unwittingly cooperates with an external attacker.
- The firewall cannot protect against the transfer of virus-infected programs or files.